Week 20: Averaging

Adult guide built from the problem files · Unpiloted; the cube sharing and the launch are unrehearsed.

This week's adult guide is week-20-facilitator.pdf, built from editable/facilitator-src/ (content.py, graphs.py and the route note). These problem files hold each problem's one-sentence idea; the full keys, hints and completeness arguments are in the guide.

Problems and routes

Problem	K–1	2–3	4–5	route
One circle between two squares	P1			first
Two circles in a row	P2			return
Squares for a 2-cube circle	P3			first
Three different cards	P4			first
A five-cube circle?	P5			return
One square controls the board	P6			return
Boards with no squares	P7			return
Change one square to raise the circle by 2		P1		first
Fill every circle at once		P2		first
A circle bigger than every square?		P3		first
Matching the biggest or smallest square		P4		first
A second filling?		P5		return
Boards with no squares		P6		return
Fill the boards and look for a second filling			P1	first
A circle of 13 or more?			P2	first
The maximum and minimum principle			P3	first
Which circles can equal the largest square?			P4	return
Two different fillings?			P5	return
Five fillings and twenty-five fillings			P6	return
Whole numbers or fractions			P7	return

First route. K–1 P1, P3–P4; Grades 2–3 P1–P4; Grades 4–5 P1–P3. **Return visits.** K–1 P2, P5–P7; Grades 2–3 P5–P6; Grades 4–5 P4–P7.

Key, one entry per problem

One circle between two squares

K–1 P1 · first route

Idea. A circle joined to two squares is their fair share: 1, 3, 2 and 4, counting an empty square as a place.

Two circles in a row

K–1 P2 · return route

Idea. Both circles must work on one unchanged board, which forces equal steps: (1,2), (2,1), (2,4) and (4,2).

Squares for a 2-cube circle

K–1 P3 · first route

Idea. Three squares from 0 to 3 that share to 2 must total 6: ten positional fillings, from (0,3,3), (1,2,3) and (2,2,2).

Three different cards

K–1 P4 · first route

Idea. With distinct cards 0-4 the middle card is halfway between the ends: eight ordered fillings, and 0 or 4 can never be the circle.

A five-cube circle?

K–1 P5 · return route

Idea. No circle can be taller than every square: the tallest stack forces its neighbours to match, all the way to a square.

One square controls the board

K–1 P6 · return route

Idea. With one square on a connected board every circle equals the square.

Boards with no squares

K–1 P7 · return route

Idea. Without squares every working board is constant: six fillings (0 to 5) on each board.

Change one square to raise the circle by 2

Grades 2–3 P1 · first route

Idea. Raising an average of d numbers by 2 needs 2d more in total, so one square must grow by 4 (two neighbours) or 6 (three).

Fill every circle at once

Grades 2–3 P2 · first route

Idea. Several circles must work on one unchanged board; paths take equal steps and the tree has the filling 4, 6.

A circle bigger than every square?

Grades 2–3 P3 · first route

Idea. The biggest number on a working board cannot sit only on circles: its neighbours must match it, all the way to a square.

Matching the biggest or smallest square

Grades 2–3 P4 · first route

Idea. A circle can equal an extreme square when a square stops the chain: equality spreads through circles, not through squares.

A second filling?

Grades 2–3 P5 · return route

Idea. Each board's filling is forced (equal steps on the path, the tree's two conditions, the diamond's two separate circles), so there is no second one.

Boards with no squares

Grades 2–3 P6 · return route

Idea. Without squares only constant fillings work: six on each board with numbers 0 to 5.

Fill the boards and look for a second filling

Grades 4–5 P1 · first route

Idea. The path (5,10), the tree (5,7) and the tied diamond (8,8) each have exactly one filling.

A circle of 13 or more?

Grades 4–5 P2 · first route

Idea. A greatest value on a finite board forces its neighbours to equal it, through circles to a square, so no circle exceeds the largest square.

The maximum and minimum principle

Grades 4–5 P3 · first route

Idea. On a finite board where every shape reaches a square, every circle lies between the smallest and largest square it can reach.

Which circles can equal the largest square?

Grades 4–5 P4 · return route

Idea. Equality with the largest square spreads through circles and stops at a square, so a square junction allows it and a circle junction does not.

Two different fillings?

Grades 4–5 P5 · return route

Idea. The difference of two fillings obeys the rule with 0 at every square, so the maximum principle makes it 0 everywhere.

Five fillings and twenty-five fillings

Grades 4–5 P6 · return route

Idea. Each connected group of circles is constant, so the triangle has 5 fillings and the two separate pairs 5 x 5 = 25.

Whole numbers or fractions

Grades 4–5 P7 · return route

Idea. Whole-number squares can force fractions: a path from A to B with L steps is whole exactly when L divides B - A.